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Department of Computer Science & Engineering

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Capstone Project Report

Group Number	G26
Title of the project	AI-Based Real-Time Trend Forecasting System
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Project domain area	Artificial Intelligence (AI)

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1. Abstract

The rapid growth of digital platforms and online information has made it increasingly difficult to identify emerging trends at an early stage. Traditional trend analysis methods often depend on manual observation and historical data, which may not effectively capture weak signals or rapidly changing trends. This project presents a **Multi-Agent AI Trend Detection and Forecasting System** designed to automatically identify, analyze, validate, and forecast emerging trends from diverse online data sources.

The proposed system uses multiple specialized agents, where each agent performs a specific task in the trend analysis process. The system collects and preprocesses trend-related data and uses a **Scout Agent** to identify potential trends and a **Weak Signal Agent** to detect early changes in trend activity. The **Lifecycle Agent** analyzes the different stages of a trend, such as Emerging, Growth, Peak, and Decline, while the **Authenticity Agent** evaluates whether a detected trend represents genuine growth or temporary hype. Further, the **Industry Impact Agent** evaluates the potential effect of the trend on different industries, and the **Financial Validation Agent** analyzes relevant business and financial factors. Machine learning techniques are used to support trend forecasting based on historical and current trend features.

Finally, a **Decision Engine** combines the outputs of the different agents to generate explainable insights and recommendations such as **Invest, Wait, or Avoid**, along with confidence scores and supporting factors. The system is integrated into a user interface called **TrendSense**, providing users with a centralized platform for monitoring and understanding emerging trends. The proposed approach aims to transform raw online data into meaningful, explainable, and actionable trend intelligence.

2. Introduction

In today's digital world, a large amount of information is generated every day through search engines, social media, websites, and other online platforms. This continuously changing information contains valuable signals about new and emerging trends. However, identifying which trends are likely to grow in the future is difficult because the data is large, unstructured, and changes rapidly.

Traditional trend analysis often depends on manual observation, historical information, and expert judgment. These approaches may fail to identify weak signals at an early stage, before a trend becomes popular. Therefore, there is a need for an intelligent system that can automatically collect and analyze online data, detect early signals, and forecast the future development of trends.

The proposed project, **Multi-Agent AI Trend Detection and Forecasting System**, addresses this problem by using multiple specialized AI agents. Each agent performs a specific task in the trend analysis process. The system includes agents for trend discovery, weak-signal detection, lifecycle analysis, authenticity validation, industry impact analysis, and financial validation. Machine learning techniques are also used to analyze historical and current trend-related features and support future trend forecasting.

The **Trend Lifecycle Agent** identifies the stage of a trend, such as Emerging, Growth, Peak, or Decline, while the **Authenticity Agent** evaluates whether a trend represents genuine growth or temporary hype. Other agents analyze the potential industry and financial impact of the identified trend. Finally, the **Decision Engine** combines the outputs of these agents to generate explainable insights and strategic recommendations such as Invest, Wait, or Avoid.

The system integrates these components into a unified platform called **TrendSense**, which provides users with a centralized interface for monitoring trends and understanding their future potential. The overall aim of the project is to transform continuously changing online data into meaningful, explainable, and actionable trend intelligence.

3. Objectives

1. To identify emerging fashion and beauty trends among Indian Gen-Z.
2. To predict short-term trend direction as UP, STABLE, or DOWN for 1-day and 3-day periods.
3. To determine whether a trend is genuine or artificial hype using multiple data sources and evidence.
4. To analyze the trend's lifecycle and market impact using ML, NLP, and multi-agent analysis.
5. To provide an explainable business recommendation such as INVEST, WAIT, or AVOID based on the prediction and supporting evidence.

4. Problem Statement

To develop a Multi-Agent AI system that can automatically detect emerging trends from diverse online data, validate their authenticity, analyze their lifecycle, forecast future behavior, and provide explainable insights about their industry and financial impact.

5. Flowchart

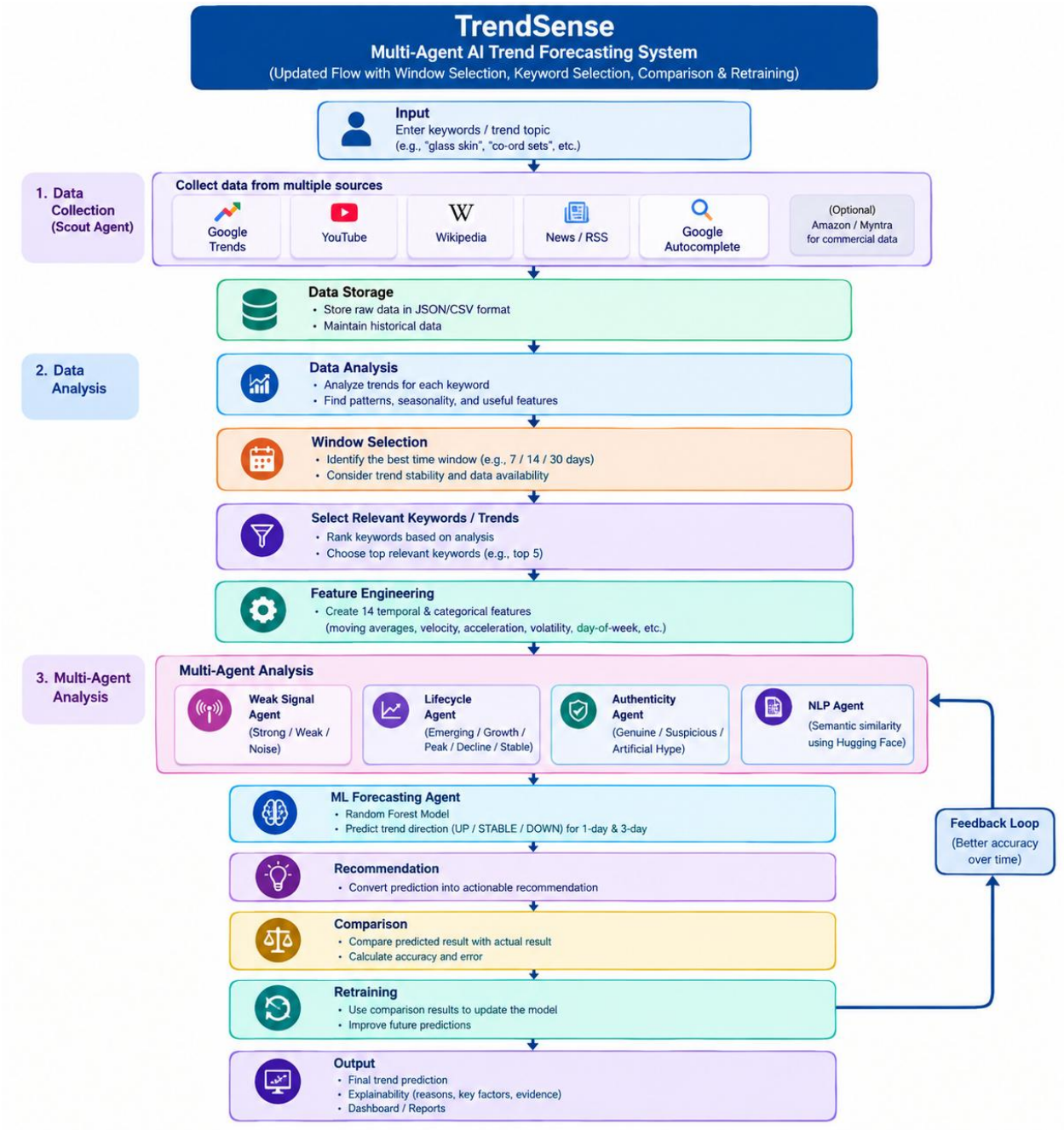


Fig. 1: Flowchart of Multi-Agent Trend Detection and Forecasting System

6. Methodology

1. Input Collection

- The system takes a set of **fashion and beauty keywords/trends** as input.
- The number of keywords is **not fixed**; the value shown in sir's diagram is only an example.

2. Data Collection

- Collect historical trend data for the given keywords from the available data sources.
- Store the collected data for further analysis.

3. Data Analysis

- Analyze the collected trend data.
- Study the **trend patterns and changes over time**.
- Identify useful information that can help in prediction.

4. Window Selection

- Select an appropriate **time window** from the historical data.
- Different windows can represent recent or longer-term trend behavior.
- The selected window is then used for further analysis and prediction.

5. Keyword/Trend Selection

- Analyze the keywords based on the selected data window.
- Identify the **most relevant keywords/trends** for prediction.
- This reduces unnecessary data and focuses the model on meaningful trends.

6. Feature Extraction

For the selected trends, useful features are generated from the time-series data, such as:

- Trend/interest values
- Moving averages
- Velocity
- Acceleration
- Volatility
- Other statistical characteristics

7. Random Forest Prediction

- The selected features are given to the **Random Forest model**.
- Random Forest analyzes multiple decision trees and combines their results.

- It predicts the future trend direction, such as:
  UP / STABLE / DOWN.

8. Recommendation

- The prediction is converted into a **useful recommendation**.
- The recommendation is based on the predicted trend and the project's decision rules.

9. Comparison

This is an important new part of sir's suggested methodology.

After some time, the system compares:

Predicted result ↔ Actual result

For example:

Predicted: UP
Actual: DOWN

The difference/error is recorded so that the model's performance can be evaluated.

10. Retraining

- The comparison results are used to improve the model.
- New/updated historical data can be added to the training process.
- The Random Forest model is retrained so that future predictions can become more accurate.

11. Final Output

The system finally provides:

- Predicted trend
- Recommendation
- Prediction performance
- Explanation/relevant information
- Updated results after retraining

Simple Methodology Flow

Input Keywords / Trends

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Data Collection

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Data Analysis

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Window Selection

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Relevant Trend Selection

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Feature Extraction

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Random Forest

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Prediction

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Recommendation

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Comparison

↓

Retraining

↓

Improved Prediction

↓

Final Output

7. Implementation

1. **Input Module**
   - Accept the required fashion/beauty keywords or trend topics.
   - The number of keywords is not fixed.
2. **Data Collection Module**
   - Collect historical trend data for the given keywords from the available sources.
   - Store the collected data for processing.
3. **Data Analysis Module**
   - Analyze the collected time-series data.
   - Identify patterns, changes, and useful trend information.
4. **Window Selection Module**
   - Select the appropriate historical time window for analysis.
   - The selected window is passed to the next stage.
5. **Relevant Trend Selection**
   - Rank/analyze the available keywords based on the selected data.
   - Select the relevant trends for further prediction.
6. **Feature Engineering**
   - Convert the selected trend data into useful numerical features such as:
     - Moving averages
     - Trend/velocity
     - Acceleration
     - Volatility
     - Other statistical features
7. **Random Forest Model**
   - Train/use the Random Forest model with the extracted features.
   - Predict the future trend direction:
     UP / STABLE / DOWN.
8. **Recommendation Module**
   - Convert the prediction into a suitable recommendation based on the project's rules.
9. **Comparison Module**
   - Compare the **predicted result with the actual result** after the prediction period.
   - Calculate the prediction error/performance.
10. **Retraining Module**
    - Use the newly available results/data to retrain the Random Forest model.
    - This creates a feedback loop for improving future predictions.
11. **Final Output**
    - Display the predicted trend, recommendation, comparison/performance results, and updated prediction.

Implementation Flow

Input

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Data Collection

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Data Analysis

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Window Selection

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Relevant Trend Selection

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Feature Engineering

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Random Forest Model

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Prediction

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Recommendation

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Compare Prediction with Actual Result

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Retrain Model

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Improved Model

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Final Output

8. Conclusion

TrendSense provides an AI-based approach for analyzing and forecasting short-term fashion and beauty trends. The system collects trend data, performs data analysis, selects a suitable time window and relevant trends, and uses Random Forest to predict whether a trend will move UP, STABLE, or DOWN.

The system also adds a comparison and retraining mechanism, where predicted results are compared with actual results and the model is updated using new data. This feedback loop helps improve future predictions.

Overall, the project aims to convert raw trend data into meaningful predictions and recommendations, while making the prediction process more systematic and explainable.

9. Future Scope

1. Automatic Trend Discovery
   The system can be extended to automatically discover new and emerging fashion/beauty keywords instead of depending only on predefined keywords.
2. Larger and More Diverse Dataset
   More historical data and a larger number of trends can be added to improve the reliability of the forecasting model.
3. Advanced Machine Learning Models
   Future versions can compare Random Forest with advanced time-series/deep-learning models such as LSTM or Temporal Fusion Transformers.
4. Improved Real-Time Analysis
   The system can be enhanced to continuously collect new data and update predictions in near real time.
5. Better Model Retraining and Feedback
   The prediction–comparison–retraining loop can be developed further so that the model learns from new outcomes and improves its future predictions.